GOLANG FOR INTERMEDIATE

ACTIVITY 1

(HTTP Request Pada Golang)

Ferdiyansyah

10121480

3KA01

File handler.go

```
package handler

import (
    "Pert4_10121480_intergolang/model" //sesuaikan dengan nama folder
(case sensitive)
    "database/sql"
    "fmt"
    "net/http"
    "strings"
)

var username, password, host, namaDB, defaultDB string
var db *sql.DB
var err error

func init() {
    username = "root"
    password = ""
    host = "localhost"
    namaDB = "db_10121480_pert4" // tulis npm masing masing
    defaultDB = "mysql"
}

func API(w http.ResponseWriter, r *http.Request) {
    db, err = model.Connect(username, password, host, namaDB)
    if err != nil {
        return
```

```
    }
    defer db.Close()

    w.Header().Set("Content-Type", "text-html; charset=utf-8;
application/json")
    dataURL := strings.Split(fmt.Sprintf("%s", r.URL.Path), "/")
    switch dataURL[2] {
    case "mahasiswa":
        switch r.Method {
        case "GET":
            HandlerMahasiswaGet(w, r)
        case "POST":
            HandlerMahasiswaPost(w, r)
        case "PUT":
            HandlerMahasiswaPut(w, r)
        case "DELETE":
            HandlerMahasiswaDelete(w, r)
        default:
            w.Write([]byte("method tidak ditemukan"))
        }
    case "nilai":
        switch r.Method {
        case "GET":
            HandlerNilaiGet(w, r)
        case "POST":
            HandlerNilaiPost(w, r)
        case "PUT":
            HandlerNilaiPut(w, r)
        case "DELETE":
            HandlerNilaiDelete(w, r)
        default:
            w.Write([]byte("method tidak ditemukan"))
        }
    case "matkul":
        switch r.Method {
        case "GET":
            HandlerMatkulGet(w, r)
        case "POST":
            HandlerMatkulPost(w, r)
        case "PUT":
```

```
            HandlerMatkulPut(w, r)
        case "DELETE":
            HandlerMatkulDelete(w, r)
        default:
            w.Write([]byte("method tidak ditemukan"))
        }
    default:
        w.Write([]byte("request tidak ditemukan"))
    }
}
```

File handleMahasiswa.go

```
package handler

import (
    "Pert4_10121480_intergolang/model" //sesuaikan dengan nama folder
(case sensitive)
    "encoding/json"
    "io/ioutil"
    "net/http"
)

func HandlerMahasiswaGet(w http.ResponseWriter, r *http.Request) {
    var data interface{}
    var err error
    npm := r.URL.Query()["npm"]
    if len(npm) != 0 {
        data, err = model.GetMahasiswa(db, npm[0])
    } else {
        data, err = model.GetAllMahasiswa(db)
    }
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
    }
    jsonData, _ := json.Marshal(data)
    w.Write(jsonData)
```

```go
}

func HandlerMahasiswaPost(w http.ResponseWriter, r *http.Request) {
    defer r.Body.Close()
    body, err := ioutil.ReadAll(r.Body)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    var data model.Mahasiswa
    if err = json.Unmarshal(body, &data); err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    if err = data.Insert(db); err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    jsonData, err := json.Marshal(data)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
    }
    w.Write(jsonData)
}

func HandlerMahasiswaDelete(w http.ResponseWriter, r *http.Request) {
    npm := r.URL.Query()["npm"]
    if len(npm) != 0 {
        data := model.Mahasiswa{NPM: npm[0]}
        if err := data.Delete(db); err != nil {
            http.Error(w, "ID tidak ditemukan", http.StatusBadRequest)
            return
        }
        w.Write([]byte("Data telah dihapus"))
    } else {
        http.Error(w, "ID tidak ditemukan",
http.StatusInternalServerError)
    }
}
```

```go
func HandlerMahasiswaPut(w http.ResponseWriter, r *http.Request) {
    npm := r.URL.Query()["npm"]
    if len(npm) == 0 {
        http.Error(w, "ID tidak ditemukan", http.StatusBadRequest)
        return
    }
    defer r.Body.Close()
    body, err := ioutil.ReadAll(r.Body)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    jsonmap := make(map[string]interface{})
    err = json.Unmarshal(body, &jsonmap)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    data := model.Mahasiswa{NPM: npm[0]}
    err = data.Update(db, jsonmap)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    result, err := model.GetMahasiswa(db, npm[0])
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    jsonData, err := json.Marshal(result)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    w.Write(jsonData)
}
```

file handleMatkul.go

```
package handler

import (
    "Pert4_10121480_intergolang/model" //isi dengan nama dan npm masing
masing
    "encoding/json"
    "io/ioutil"
    "net/http"
)

func HandlerMatkulGet(w http.ResponseWriter, r *http.Request) {
    var data interface{}
    var err error
    kd_mk := r.URL.Query()["kd_mk"]
    if len(kd_mk) != 0 {
        data, err = model.GetMatkul(db, kd_mk[0])
    } else {
        data, err = model.GetAllMatkul(db)
    }
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
    }
    jsonData, _ := json.Marshal(data)
    w.Write(jsonData)
}

func HandlerMatkulPost(w http.ResponseWriter, r *http.Request) {
    defer r.Body.Close()
    body, err := ioutil.ReadAll(r.Body)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    var data model.Matkul
    if err = json.Unmarshal(body, &data); err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    if err = data.Insert(db); err != nil {
```

```go
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    jsonData, err := json.Marshal(data)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
    }
    w.Write(jsonData)
}

func HandlerMatkulDelete(w http.ResponseWriter, r *http.Request) {
    kd_mk := r.URL.Query()["kd_mk"]
    if len(kd_mk) != 0 {
        data := model.Matkul{Kd_mk: kd_mk[0]}
        if err := data.Delete(db); err != nil {
            http.Error(w, "ID tidak ditemukan", http.StatusBadRequest)
            return
        }
        w.Write([]byte("Data telah dihapus"))
    } else {
        http.Error(w, "ID tidak ditemukan",
http.StatusInternalServerError)
    }
}

func HandlerMatkulPut(w http.ResponseWriter, r *http.Request) {
    kd_mk := r.URL.Query()["kd_mk"]
    if len(kd_mk) == 0 {
        http.Error(w, "ID tidak ditemukan", http.StatusBadRequest)
        return
    }
    defer r.Body.Close()
    body, err := ioutil.ReadAll(r.Body)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    jsonmap := make(map[string]interface{})
    err = json.Unmarshal(body, &jsonmap)
    if err != nil {
```

```
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    data := model.Matkul{Kd_mk: kd_mk[0]}
    err = data.Update(db, jsonmap)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    result, err := model.GetMatkul(db, kd_mk[0])
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    jsonData, err := json.Marshal(result)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    w.Write(jsonData)
}
```

file handleNilai.go

```
package handler

import (
    "Pert4_10121480_intergolang/model" //sesuaikan dengan nama folder
(case sensitive)
    "encoding/json"
    "io/ioutil"
    "net/http"
)

func HandlerNilaiGet(w http.ResponseWriter, r *http.Request) {
    var data interface{}
    var err error
```

```go
    npm := r.URL.Query()["npm"]
    if len(npm) != 0 {
        data, err = model.GetNilai(db, npm[0])
    } else {
        data, err = model.GetAllNilai(db)
    }
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
    }
    jsonData, _ := json.Marshal(data)
    w.Write(jsonData)
}

func HandlerNilaiPost(w http.ResponseWriter, r *http.Request) {
    defer r.Body.Close()
    body, err := ioutil.ReadAll(r.Body)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    var data model.Nilai
    if err = json.Unmarshal(body, &data); err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    if err = data.Insert(db); err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    jsonData, err := json.Marshal(data)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
    }
    w.Write(jsonData)
}

func HandlerNilaiDelete(w http.ResponseWriter, r *http.Request) {
    npm := r.URL.Query()["npm"]
    if len(npm) != 0 {
        data := model.Nilai{NPM: npm[0]}
```

```go
        if err := data.Delete(db); err != nil {
            http.Error(w, "ID tidak ditemukan", http.StatusBadRequest)
            return
        }
        w.Write([]byte("Data telah dihapus"))
    } else {
        http.Error(w, "ID tidak ditemukan",
http.StatusInternalServerError)
    }
}

func HandlerNilaiPut(w http.ResponseWriter, r *http.Request) {
    npm := r.URL.Query()["npm"]
    if len(npm) == 0 {
        http.Error(w, "ID tidak ditemukan", http.StatusBadRequest)
        return
    }
    defer r.Body.Close()
    body, err := ioutil.ReadAll(r.Body)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    jsonmap := make(map[string]interface{})
    err = json.Unmarshal(body, &jsonmap)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    data := model.Nilai{NPM: npm[0]}
    err = data.Update(db, jsonmap)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    result, err := model.GetNilai(db, npm[0])
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
```

```go
    jsonData, err := json.Marshal(result)
    if err != nil {
        http.Error(w, err.Error(), http.StatusInternalServerError)
        return
    }
    w.Write(jsonData)
}
```

file main.go

```go
package main

import (
    "Pert4_10121480_intergolang/handler" //sesuaikan dengan nama folder
(case sensitive)
    "log"
    "net/http"
)

func main() {
    http.HandleFunc("/api/", handler.API)
    //Ganti 2 digit akhir port dengan 2 digit akhir NPM anda

    // Karena digit belakang npm saya 80, dan port 8080 sudah terpakai
    // maka saya menggunakan 2 digit sebelum terakhir yaitu 48
    log.Println("localhost : 8048")
    http.ListenAndServe(":8048", nil)
}
```

Ouput Terminal Test

```
C:\Golang\src\Pert4_10121480_intergolang>cd test

C:\Golang\src\Pert4_10121480_intergolang\test>go test -run TestDatabase
PASS
ok      Pert4_10121480_intergolang/test 0.039s
```

```
C:\Golang\src\Pert4_10121480_intergolang\test>go test -run TestMahasiswa
&{12345678 Abdi Teh Ayeuna 3KA20}
&{19283746 Doremi Budi 4KA20}
&{44444444 DoBud 4KA21}
PASS
ok      Pert4_10121480_intergolang/test 0.074s
```

```
C:\Golang\src\Pert4_10121480_intergolang\test>go test -run TestMatkul
&{KA01 Algoritma 1}
PASS
ok      Pert4_10121480_intergolang/test 0.059s
```

```
C:\Golang\src\Pert4_10121480_intergolang\test>go test -run TestNilai
&{1 KA02 19283746 30 30 30 E}
PASS
ok      Pert4_10121480_intergolang/test 0.062s
```

Phpmyadmin

Tabel	Tindakan						Baris	Jenis	Penyortiran	Ukuran	Beban
mahasiswa	Jelajahi	Struktur	Cari	Tambahkan	Kosongkan	Hapus	0	InnoDB	utf8_general_ci	16.0 KB	-
matkul	Jelajahi	Struktur	Cari	Tambahkan	Kosongkan	Hapus	0	InnoDB	utf8_general_ci	16.0 KB	-
nilai	Jelajahi	Struktur	Cari	Tambahkan	Kosongkan	Hapus	1	InnoDB	utf8_general_ci	16.0 KB	-
3 tabel	Jumlah						1	InnoDB	utf8_general_ci	48.0 KB	0 B

Ouput Terminal main.go

```
C:\Golang\src\Pert4_10121480_intergolang>go run main.go
2023/11/11 14:20:28 localhost : 8048
```

Ouput Api GET Mahasiswa

localhost:8048/api/mahasiswa

[{"NPM":"19283746","Nama":"Doremi Budi","Kelas":"4KA20"},{"NPM":"44444444","Nama":"DoBud","Kelas":"4KA21"}]

Ouput Api GET Matkul

localhost:8048/api/matkul

[{"Kd_mk":"KA02","Matakuliah":"B.Indo Dasar"},{"Kd_mk":"KA3","Matakuliah":"IPA Dasar"}]

Ouput Api GET Nilai

localhost:8048/api/nilai

[{"id_nilai":2,"kd_mk":"KA03","npm":"44444444","uas":50,"uts":50,"total":50,"grade":"D"}]

NOTE: NPM saya 10121480, dua digit terakhirnya 80. Karena port 8080 sudah digunakan di komputer ini maka saya ubah menjadi 48.

```go
package main

import (
    "Pert4_10121480_intergolang/handler" //sesuaikan dengan nama fo
    "log"
    "net/http"
)

func main() {
    http.HandleFunc("/api/", handler.API)
    //Ganti 2 digit akhir port dengan 2 digit akhir NPM anda

    // Karena digit belakang npm saya 80, dan port 8080 sudah terpa
    // maka saya menggunakan 2 digit sebelum terakhir yaitu 48
    log.Println("localhost : 8080")
    http.ListenAndServe(":8080", nil)
}
```

```
C:\Golang\src\Pert4_10121480_intergolang>go run
2023/11/11 14:16:50 localhost : 8085
exit status 2

C:\Golang\src\Pert4_10121480_intergolang>go run
2023/11/11 14:18:10 localhost : 8048
&{19283746 Doremi Budi 4KA20}
&{44444444 DoBud 4KA21}
exit status 2

C:\Golang\src\Pert4_10121480_intergolang>cd tes

C:\Golang\src\Pert4_10121480_intergolang\test>g
&{KA01 Algoritma 1}
PASS
ok      Pert4_10121480_intergolang/test 0.059s

C:\Golang\src\Pert4_10121480_intergolang\test>c

C:\Golang\src\Pert4_10121480_intergolang>go run
2023/11/11 14:20:28 localhost : 8048
exit status 2

C:\Golang\src\Pert4_10121480_intergolang>go tes
?       Pert4_10121480_intergolang      [no tes

C:\Golang\src\Pert4_10121480_intergolang>go run
2023/11/11 14:34:55 localhost : 8048
&{19283746 Doremi Budi 4KA20}
&{44444444 DoBud 4KA21}
exit status 2

C:\Golang\src\Pert4_10121480_intergolang>go run
2023/11/11 14:38:01 localhost : 8080

C:\Golang\src\Pert4_10121480_intergolang>
```